Rational Number Project

Initial Fraction Ideas Lesson 4: Overview	Materials
Students use paper folding to model and name unit and non-unit fractions. Students compare the paper-folding model to fraction circles. Students record fractions in words: one-fourth, two-thirds.	• Paper strips for folding for students • Fraction Circles for teacher • Student Pages A-L

Teaching Actions

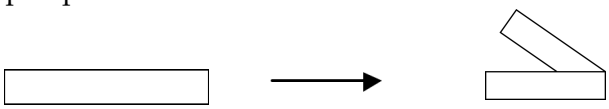
Warm Up

Name the red piece in three different ways by changing the unit. What different units did you use?

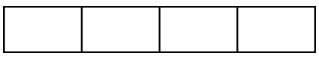
Large Group Introduction

1. Prior to using paper strips to model fractions it is necessary to practice folding strips into 2, 3, 4, 6, and 8 equal parts.

   Ask students to follow along with you as you model how to fold paper strips. Fold paper strip into two equal parts:



2. Keep it folded. Now fold it again into two equal parts. Ask: how many equal parts do you think we have? Unfold:



3. Ask students to verbalize how to fold paper strips to form four equal parts.

4. Model folding into three equal parts. Form the letter "S" with a paper strip to get close to 3 equal parts. Press down on paper.

5. Model sixths. Fold paper strip into thirds and then fold into two equal parts. Have students do this and

Comments

Cut paper strips from 8.5" by 11" sheets of paper about 1 inch wide and 8.5" long.

This lesson should take two class periods. Students are still recording fractional amounts using word names; symbols are introduced in lesson 5.

One way to break down the lesson is to complete steps 1 to 6 where students learn to fold paper into 2, 3, 4, 6, and 8 equal parts; then have students practice on their own using Student Page A. Wrap up the lesson with students sharing how they folded paper strips. Select students you have seen who are successful so they can model for others how to fold paper strips into equal parts.

Then pick up the lesson at Step 7 starting with the second warm up and continue on in lesson where students shade equal parts of the paper strips to show fraction amounts. Here students compare the paper strip model with fraction circles. Students can do selected student pages for practice and homework.

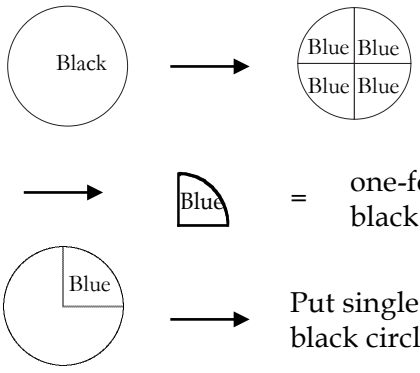
Folding into thirds is tricky. For some students you may want to use precut strips with the partitioning lines drawn in. (Student Page L)

Teaching Actions

guess, before unfolding, the number of equal parts they expect.

6. Ask students if they could have obtained sixths by folding first in halves and then in thirds? Try it.

7. Now that students have had some practice folding paper strips, they can shade equal parts of paper strips to show fractions. Using fraction circles, show one-fourth using a black circle as the unit.



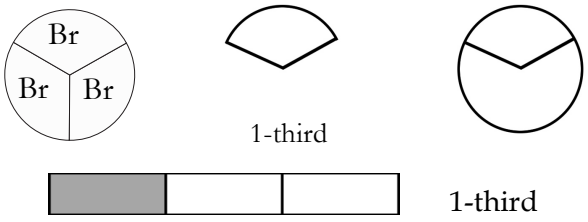
Say: To show one-fourth of a black circle I divided it into four equal parts. Pick up one of the parts to show one-fourth.

9. Ask: How can you show me one-fourth with a paper strip? Have students fold into 4 equal parts and shade in one of the 4 equal parts. Record fraction name as 1-fourth.

10. Discuss how the two displays for one-fourth are alike and different.

11. Repeat for 1-third and 1-eighth

12. Look at two displays for one-third:



Comments

Students often will expect 5 equal parts (3+2). They are more apt to think additively than multiplicatively.

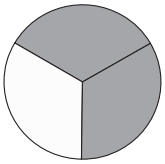
The similarity between the two displays is what's important. A unit or whole is divided into <u>equal</u> parts and one or more equal parts are highlighted in some way. This is a *manipulative to manipulative* translation.

Teaching Actions

13. Shade in another third on the paper strip.



Ask: how many thirds are shown now? How can I show two-thirds with circles? (Pick up two browns and say these are two-thirds of black.) State that 2-thirds is 1-third and 1-third more:



14. Now draw a picture of a square. Divide it into 4 equal parts and shade 3 of 4 parts. Ask students to fold paper to show the same fraction that you drew. Record fractions as 3-fourths: 1-fourth + 1-fourth + 1-fourth.

15. Return to fraction circles. Model problems as in lesson 3, this time with non-unit fractions.

Examples:

- The black circle = 1. What is the value of 1 blue; 3 blues; 1 brown; 2 browns; 3 reds.
- The yellow piece = 1. What is the value of 1 blue; 2 reds; 3 grays; 2 pinks.

Small Group/Partner Work

16. There are several student pages in this lesson. **Select the most appropriate ones for your students.** Students may need some assistance to do some of the pages. See Comments for clarification.

Wrap Up

17. Ask students to identify problems on Student Pages, H, I & J, that equal zero. Ask: How can you show 0 using paper strips? Ask students to do this in more than one way. (For example: 0/2; 0/4; 0/3)

Comments

Non-unit fractions are introduced as sums of unit fractions: 2-fourths is 1-fourth and 1-fourth.

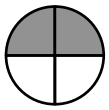
Students now have seen two models for fractions. Practice pages that follow this lesson give students a chance to apply their new learning to pictures of units in different shapes.

Teacher Notes for Student Pages:

E: Clarify with students that a picture may need to be modified to determine if 2-fourths are shaded in. For example:



Is 2-fourths shaded?



2-fourths can easily be seen once the picture is completed by drawing in the needed lines.

Teaching Actions	Comments
18. Ask: What are the different ways to model one and zero with paper folding strips?	

Translations

- Manipulative to verbal
- Manipulative to manipulative to verbal
- Manipulative to verbal to written symbols (word names)

Name the red piece in three different ways by changing the unit. What different units did you use?

Fold a paper strip into 6 equal parts. Explain to a friend how you did this.

Did you both do it in the same way?

Practice Folding Paper

You and your teacher folded paper strips into different numbers of equal parts. Now it is your turn to practice paper folding on your own. You will need at least 4 paper strips.

1. Imagine that one of these paper strips is a candy bar. You want to share the candy bar evenly so you and three friends get the same amount. Show how to do that with the paper strip and then draw a picture of it below.

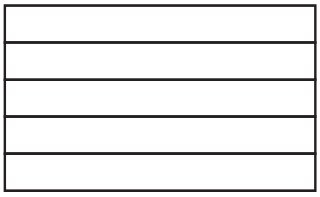
2. Take out one of the paper strips and try to fold it into 3 equal parts. Describe below how you did this.

3. How can you partition the paper strip into 6 equal parts? Before you do this, write up the steps below. Then act out those steps to fold the paper strip into 6 equal parts.

4. Fold a paper strip into 8 equal parts. Then draw a picture of the strip below.

For each problem, look at the figure and then answer the question and write the word name for each fraction. Word names may include: half, halves, third, fourth, fifth, sixth, eighth or tenth.

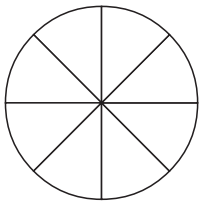
1.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

2.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

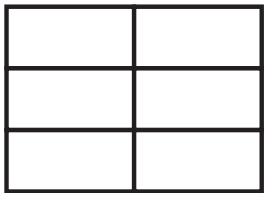
3.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

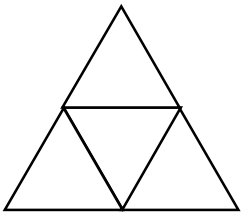
4.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

5.

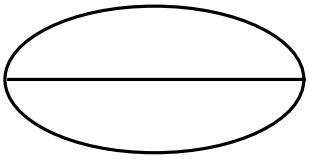


_____ equal-sized parts.

Each part is _____ _____ of the whole.

Name_____ Grade 3 Lesson 4 / Student Page C

6.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

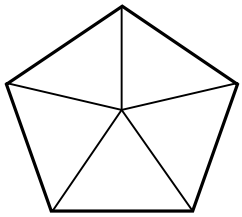
7.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

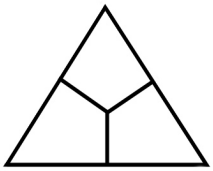
8.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

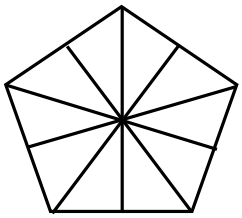
9.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

10.



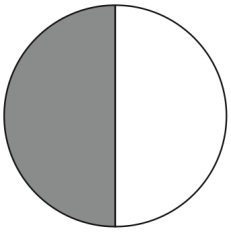
_____ equal-sized parts.

Each part is _____ _____ of the whole.

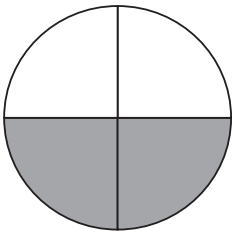
Directions:

You'll need paper strips for folding. For any four of the figures shown below, fold paper strips to model the fraction that the figure models. After you have folded and shaded your paper, write on it the fraction you have shown (use words, not symbols).

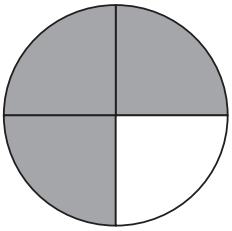
1.



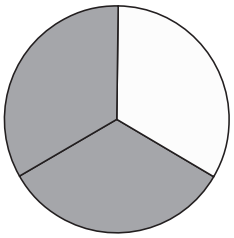
2.



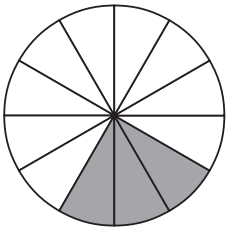
3.



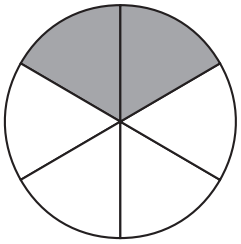
4.



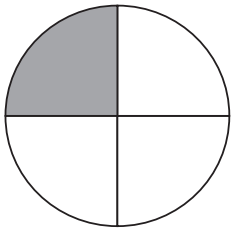
5.



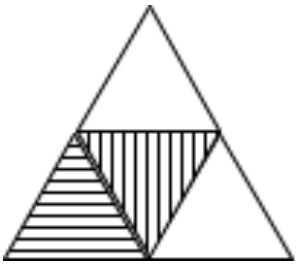
6.



7.



8.



Directions:

You'll need paper strips for folding. For any four of the figures shown below, fold paper strips to model the fraction that the figure models. After you have folded and shaded your paper, write on it the fraction you have shown (use words, not symbols).

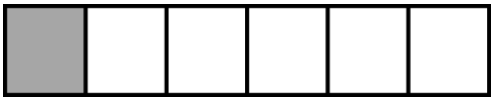
1.



2.



3.



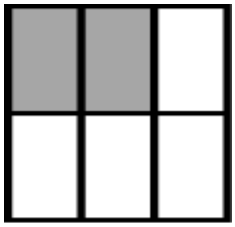
4.



5.



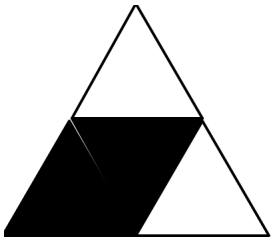
6.



7.

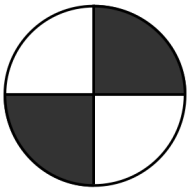
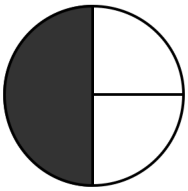
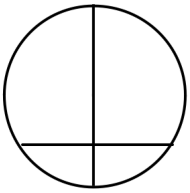
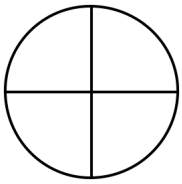
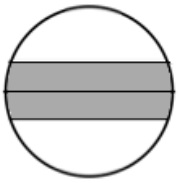


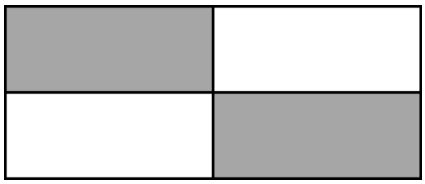
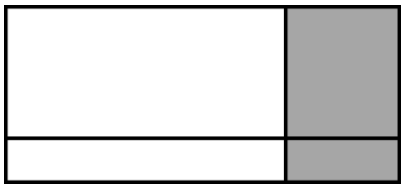
8.



Name_____ Grade 3 Lesson 4 / Student Page F

Look at each picture carefully. Place an "X" beside each picture that shows 2-fourths shaded in. You may need to draw in lines to determine if 2-fourths are shaded.





For each diagram, fill in the blanks to tell about the diagram.

a.



Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____ -sixth

b.



Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is 1- _____

c.

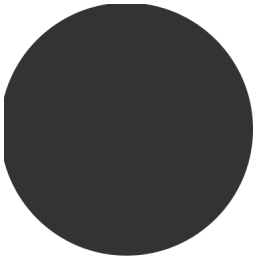


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

d.

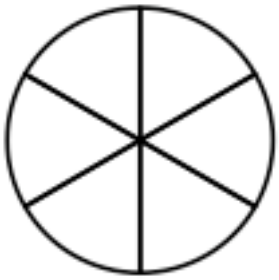


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

e.

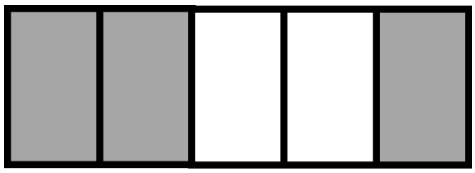


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

f.



Number of equal parts _____

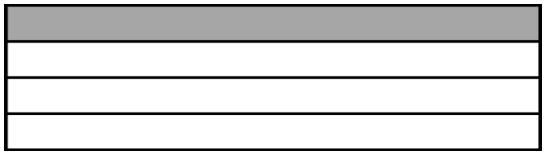
Number of equal parts shaded _____

The fraction shaded is _____

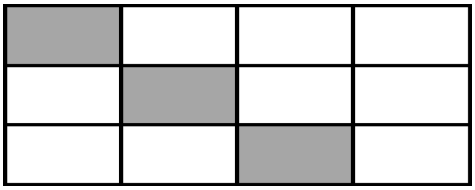
Write words like 2-fourths, 3-fifths, and so on for the fraction shaded by each diagram.



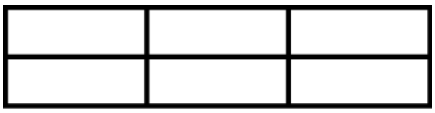
Write _____



Write _____



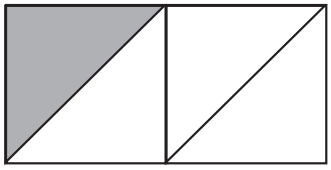
Write _____



Write _____



Write _____



Write _____

Write the fraction that is shown in words:

a.



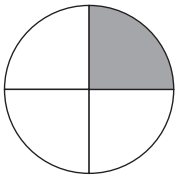
_____

b.



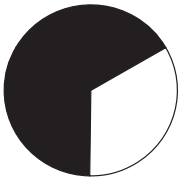
_____

c.



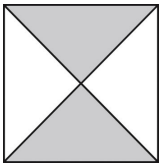
_____

d.



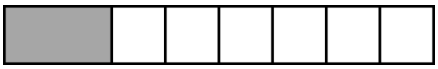
_____

e.

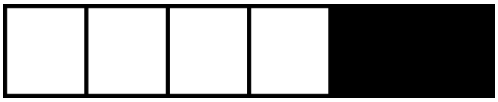


_____

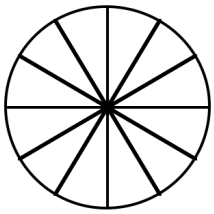
f.



_____

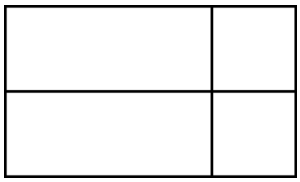
g. 

_____

h. 

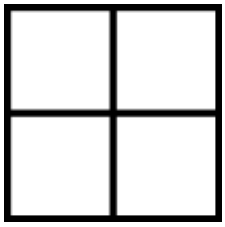
_____

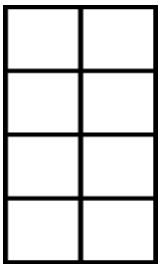
Circle the figures that have equal-sized parts.

1. 

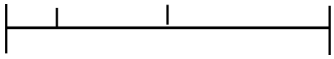
2. 

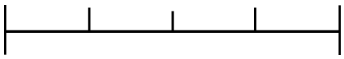
3. 

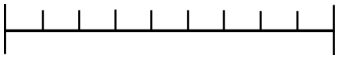
4. 

5. 

6. 

7. 

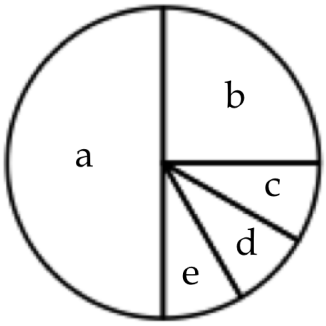
8. 

9. 

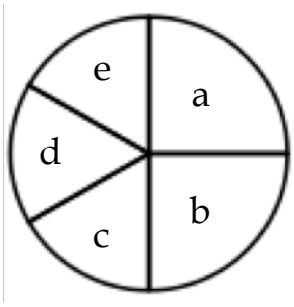
Problem Solving

Directions:

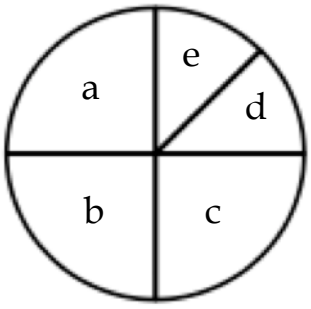
For each of the drawings write the color corresponding to the part marked a, b, c, and so on. Then write the word name for the fraction that the color represents. You can use fraction circles if you need them. Your teacher will help you with exercise 1.



Color	Fractions in Words
a. yellow	1-half
b. blue	1-fourth
c.	
d.	
e.	



Color	Fractions in Words
a.	
b.	
c.	
d.	
e.	



Color	Fractions in Words
a.	
b.	
c.	
d.	
e.	